WEATHER ELEMENTS / INDIA JET STREAM-WESTERN DISTURBANCES

WEATHER CAUSATION CHAIN → HEAT-BUDGET LEDGER → STABILITY PROFILES → WIND-FORCE TRIANGLE → PLANETARY CIRCULATION RAIL → HUMIDITY-TO-CLOUD CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 WEATHER CAUSATION CHAIN

WEATHER CAUSATION CHAIN | SURFACE AND ATMOSPHERIC HEATING | TEMPERATURE GRADIENT | PRESSURE GRADIENT | MOISTURE TRANSPORT AND CONVERGENCE | LIFT + SATURATION | CLOUD AND PRECIPITATION | MUST REMEMBER

INSOLATION → surface and atmospheric heating → **TEMPERATURE GRADIENT → pressure gradient** → **WIND → moisture transport and convergence** → **LIFT + SATURATION → cloud and precipitation** → **MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient force, Coriolis/friction, humidity, stability, condensation, clouds**

CORE CLAIM
- INSOLATION → surface and atmospheric heating
- TEMPERATURE GRADIENT → pressure gradient

MECHANISM
- WIND → moisture transport and convergence
- LIFT + SATURATION → cloud and precipitation

CONSEQUENCE / LIMIT
- MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient force, Coriolis/friction, humidity, stability, condensation, clouds and precipitation
- upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian seasonality.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian seasonality.

01

STAGE 01 HEAT-BUDGET LEDGER

HEAT-BUDGET LEDGER | INCOMING SHORTWAVE | REFLECTED + ABSORBED ENERGY | LONGWAVE + SENSIBLE + LATENT HEAT | TEMPERATURE AND CIRCULATION RESPONSE

SUN → incoming shortwave → **SURFACE → reflected + absorbed energy** → **EARTH → longwave + sensible + latent heat** → **BALANCE → temperature and circulation response**

CORE CLAIM
- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

MECHANISM
- EARTH → longwave + sensible + latent heat
- BALANCE → temperature and circulation response

CONSEQUENCE / LIMIT
- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARTH → longwave + sensible + latent heat.

02

STAGE 02 STABILITY PROFILES

STABILITY PROFILES | NORMAL TENDENCY | TEMPERATURE FALLS UPWARD | TEMPERATURE RISES THROUGH LAYER | STABLE LAYER | VERTICAL MIXING SUPPRESSED | SURFACE INVERSION | FOG AND POLLUTION TRAPPING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NORMAL TENDENCY → temperature falls upward	STABLE LAYER → vertical mixing suppressed	NORMAL TENDENCY → temperature falls upward
INVERSION → temperature rises through layer	SURFACE INVERSION → fog and pollution trapping	INVERSION → temperature rises through layer

CORE CLAIM
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

MECHANISM
- STABLE LAYER → vertical mixing suppressed
- SURFACE INVERSION → fog and pollution trapping

CONSEQUENCE / LIMIT
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

03

STAGE 03 WIND-FORCE TRIANGLE

WIND-FORCE TRIANGLE | PRESSURE GRADIENT | STARTS ACCELERATION | RIGHT NORTH | LEFT SOUTH | SLOWS AND CROSSES ISOBARS | NEAR-GEOSTROPHIC PARALLEL FLOW

PRESSURE GRADIENT → starts acceleration → **CORIOLIS → right north / left south** → **FRICTION → slows and crosses isobars** → **ALOFT → near-geostrophic parallel flow**

CORE CLAIM
- PRESSURE GRADIENT → starts acceleration

MECHANISM
- FRICTION → slows and crosses isobars

CONSEQUENCE / LIMIT
- PRESSURE GRADIENT → starts acceleration

CORE CLAIM
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

MECHANISM
- STABLE LAYER → vertical mixing suppressed
- SURFACE INVERSION → fog and pollution trapping

CONSEQUENCE / LIMIT
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

03

STAGE 03 WIND-FORCE TRIANGLE

WIND-FORCE TRIANGLE · PRESSURE GRADIENT · STARTS ACCELERATION · RIGHT NORTH · LEFT SOUTH · SLOWS AND CROSSES ISOBARS · NEAR-GEOSTROPHIC PARALLEL FLOW

PRESSURE GRADIENT → starts acceleration → **CORIOLIS → right north / left south** → **FRICTION → slows and crosses isobars** → **ALOFT → near-geostrophic parallel flow**

CORE CLAIM
- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

MECHANISM
- FRICTION → slows and crosses isobars
- ALOFT → near-geostrophic parallel flow

CONSEQUENCE / LIMIT
- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRICTION → slows and crosses isobars; ALOFT → near-geostrophic parallel flow.

04

STAGE 04 PLANETARY CIRCULATION RAIL

PLANETARY CIRCULATION RAIL · EQUATORIAL LOW · CONVERGENCE AND ASCENT · SUBTROPICAL HIGH · DESCENT AND TRADES · SUBPOLAR LOW · WESTERLY-POLAR INTERACTION · POLAR HIGH

EQUATORIAL LOW → convergence and ascent

SUBTROPICAL HIGH → descent and trades → **SUBPOLAR LOW → westerly-polar interaction** → **POLAR HIGH → cold outflow; belts migrate**

CORE CLAIM
- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

MECHANISM
- SUBPOLAR LOW → westerly-polar interaction
- POLAR HIGH → cold outflow; belts migrate

CONSEQUENCE / LIMIT
- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUBPOLAR LOW → westerly-polar interaction; POLAR HIGH → cold outflow; belts migrate.

05

STAGE 05 HUMIDITY-TO-CLOUD CHAIN

HUMIDITY-TO-CLOUD CHAIN · MOISTURE CONTENT + TEMPERATURE · RELATIVE HUMIDITY · COOLING TO SATURATION · DEW POINT REACHED · NUCLEI + LIFT · DROPLETS OR ICE CRYSTALS · GROWTH + FALL SPEED

MOISTURE CONTENT + TEMPERATURE → relative humidity → **COOLING TO SATURATION → dew point reached** → **NUCLEI + LIFT → droplets or ice crystals** → **GROWTH + FALL SPEED → precipitation**

CORE CLAIM
- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

MECHANISM
- NUCLEI + LIFT → droplets or ice crystals
- GROWTH + FALL SPEED → precipitation

CONSEQUENCE / LIMIT
- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

06

STAGE 06 RAINFALL COMPARISON

RAINFALL COMPARISON · BUOYANT THERMAL ASCENT · FORCED RELIEF ASCENT · CONVERGENT DYNAMIC ASCENT · REAL EVENT · MECHANISMS MAY COMBINE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CONVECTIONAL → buoyant thermal ascent	FRONTAL / CYCLONIC → convergent dynamic ascent	CONVECTIONAL → buoyant thermal ascent
OROGRAPHIC → forced relief ascent	REAL EVENT → mechanisms may combine	OROGRAPHIC → forced relief ascent

CORE CLAIM
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

MECHANISM
- FRONTAL / CYCLONIC → convergent dynamic ascent
- REAL EVENT → mechanisms may combine

CONSEQUENCE / LIMIT
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

06

STAGE 06

RAINFALL COMPARISON

RAINFALL COMPARISON · BUOYANT THERMAL ASCENT · FORCED RELIEF ASCENT · CONVERGENT DYNAMIC ASCENT · REAL EVENT · MECHANISMS MAY COMBINE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CONVECTIONAL → buoyant thermal ascent	FRONTAL / CYCLONIC → convergent dynamic ascent	CONVECTIONAL → buoyant thermal ascent
OROGRAPHIC → forced relief ascent	REAL EVENT → mechanisms may combine	OROGRAPHIC → forced relief ascent

CORE CLAIM
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

MECHANISM
- FRONTAL / CYCLONIC → convergent dynamic ascent
- REAL EVENT → mechanisms may combine

CONSEQUENCE / LIMIT
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

07

STAGE 07

TROPOSPHERE SECTION

TROPOSPHERE SECTION · MOISTURE, FRICTION AND HEATING · MID LEVELS · CLOUDS, TROUGHS AND STABILITY · UPPER TROPOSPHERE · JETS AND DIVERGENCE · TROPICS DEEPER · POLES SHALLOWER

SURFACE → moisture, friction and heating

MID LEVELS → clouds, troughs and stability → UPPER TROPOSPHERE → jets and divergence → TROPICS DEEPER / POLES SHALLOWER

CORE CLAIM
- SURFACE → moisture, friction and heating
- MID LEVELS → clouds, troughs and stability

MECHANISM
- UPPER TROPOSPHERE → jets and divergence
- TROPICS DEEPER / POLES SHALLOWER

CONSEQUENCE / LIMIT
- SURFACE → moisture, friction and heating
- MID LEVELS → clouds, troughs and stability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPPER TROPOSPHERE → jets and divergence; TROPICS DEEPER / POLES SHALLOWER.

08

STAGE 08

JET-STREAM ENGINE

JET-STREAM ENGINE · HORIZONTAL TEMPERATURE GRADIENT · PRESSURE GRADIENT ALOFT · CORIOLIS BALANCE · STRONG NEAR-ZONAL WIND · ROSSBY WAVES · RIDGES, TROUGHS AND STEERING

HORIZONTAL TEMPERATURE GRADIENT → HEIGHT / PRESSURE GRADIENT ALOFT → CORIOLIS BALANCE → strong near-zonal wind → ROSSBY WAVES → ridges, troughs and steering

CORE CLAIM
- HORIZONTAL TEMPERATURE GRADIENT
- HEIGHT / PRESSURE GRADIENT ALOFT

MECHANISM
- CORIOLIS BALANCE → strong near-zonal wind
- ROSSBY WAVES → ridges, troughs and steering

CONSEQUENCE / LIMIT
- HORIZONTAL TEMPERATURE GRADIENT
- HEIGHT / PRESSURE GRADIENT ALOFT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORIOLIS BALANCE → strong near-zonal wind; ROSSBY WAVES → ridges, troughs and steering.

09

STAGE 09

INDIA SEASONAL JETS

INDIA SEASONAL JETS · STWJ SOUTH OF HIMALAYA SECTOR · STEERS WESTERN DISTURBANCES EAST · TROPICAL EASTERLY JET DEVELOPS · JETS ARE PARTS OF WIDER CIRCULATION · CLOSE DISTINCTION · HUMIDITY, RELATIVE HUMIDITY AND DEW POINT DIFFER; WEATHER DIFFERS

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east → SUMMER → tropical easterly jet develops → RULE → jets are parts of wider circulation

CORE CLAIM
- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

MECHANISM
- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Humidity, relative humidity and dew point differ
- weather differs from climate, jet stream from surface wind, and a western disturbance is an eastward-moving extratropical system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

09

STAGE 09 INDIA SEASONAL JETS

INDIA SEASONAL JETS | STWJ SOUTH OF HIMALAYA SECTOR | STEERS WESTERN DISTURBANCES EAST | TROPICAL EASTERLY JET DEVELOPS | JETS ARE PARTS OF WIDER CIRCULATION | CLOSE DISTINCTION | HUMIDITY, RELATIVE HUMIDITY AND DEW POINT DIFFER; WEATHER DIFFERS

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east → SUMMER → tropical easterly jet develops → RULE → jets are parts of wider circulation

CORE CLAIM

- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

MECHANISM

- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Humidity, relative humidity and dew point differ
- weather differs from climate, jet stream from surface wind, and a western disturbance is an eastward-moving extratropical system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

10

STAGE 10 WESTERN-DISTURBANCE CHAIN

WESTERN-DISTURBANCE CHAIN | UPPER TROUGH | EASTWARD WESTERLY MOTION | MOISTURE + DYNAMICAL LIFT | CLOUD AND RAIN/SNOW | HIMALAYA + PLAINS | TERRAIN-SPECIFIC IMPACTS | BENEFICIAL OR DAMAGING BY TIMING/INTENSITY

UPPER TROUGH / LOW → eastward westerly motion → MOISTURE + DYNAMICAL LIFT → cloud and rain/snow → HIMALAYA + PLAINS → terrain-specific impacts → OUTCOME → beneficial or damaging by timing/intensity → EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution

CORE CLAIM

- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

MECHANISM

- HIMALAYA + PLAINS → terrain-specific impacts
- OUTCOME → beneficial or damaging by timing/intensity

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution
- event attribution and seasonal forecasts require dated agency evidence and probabilistic language.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution; event attribution and seasonal forecasts require dated agency evidence and probabilistic language.

11

STAGE 11 FORECAST EVIDENCE LADDER

FORECAST EVIDENCE LADDER | PILOT BALLOON | UPPER-AIR STATE | SATELLITE + RADAR | CLOUD AND PRECIPITATION | NWP ENSEMBLE | SCENARIOS AND UNCERTAINTY | ACTION; LATER OBSERVATIONS VERIFY OUTCOME

RADIOSONDE / PILOT BALLOON → upper-air state → SATELLITE + RADAR → cloud and precipitation → NWP ENSEMBLE → scenarios and uncertainty → WARNING → action; later observations verify outcome

CORE CLAIM

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

MECHANISM

- NWP ENSEMBLE → scenarios and uncertainty
- WARNING → action; later observations verify outcome

CONSEQUENCE / LIMIT

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NWP ENSEMBLE → scenarios and uncertainty; WARNING → action; later observations verify outcome.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | JET STREAMS ARE FAST, NARROW UPPER-TROPOSPHERIC WIND BANDS. | STWJ LIES SOUTH OF THE HIMALAYAS IN WINTER AND | NORTHWARD WITHDRAWAL OF STWJ IS A PRE-CONDITION FOR SW | TEJ DEVELOPS AS PART OF THE SUMMER MONSOON'S UPPER-AIR | WESTERN DISTURBANCES ARE EXTRA-TROPICAL, NON-MONSOONAL LOW-PRESSURE SYSTEMS FROM THE | WDS BRING WINTER/EARLY-SPRING RAIN TO PUNJAB, HARYANA, DELHI AND | THEY ARE VITAL FOR RABI CROPS SUCH AS WHEAT

OPTIONAL SOURCE DEPTH

- Jet streams are fast, narrow upper-tropospheric wind bands.
- STWJ lies south of the Himalayas in winter and acts as the conveyor for Western Disturbances.
- Northward withdrawal of STWJ is a pre-condition for SW monsoon onset.
- TEJ develops as part of the summer monsoon's upper-air circulation. Tibetan/Asian heating matters,

ADVANCED DEBATE

- Western Disturbances are extra-tropical, non-monsoonal low-pressure systems from the Mediterranean moving east via Iran-Afghanistan-Pakistan.
- WDs bring winter/early-spring rain to Punjab, Haryana, Delhi and UP and snow to J&K, HP and Uttarakhand
- they are vital for Rabi crops such as wheat, mustard and gram.
- In May–June, intense heating over north-west India supports the hot, dry Loo. Its timing and strength

LEGACY / NUANCE

- Altitude lowers temperature by about 6.5 deg C/km in the notes
- the Himalayan barrier gives India warmer winters than comparable latitudes by blocking Central Asian cold air.
- STWJ steers Western Disturbances in winter.
- STWJ's northward shift is essential for SW monsoon onset.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels